

AYMANE AIT DADS

Intern Software Engineer - Edge AI Engineering | Model Deployment · ML Pipeline Engineering · Edge Inference Optimization

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PROFESSIONAL SUMMARY

End-to-end ML pipeline deployed on 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by ~60% through automated feature engineering, clustering, and stakeholder-ready Power BI outputs - directly demonstrating production-grade AI deployment capability. Fine-tuned a 30B Mamba-Transformer MoE model on a Kaggle competition with a \$106K+ prize pool, reaching top 10% public leaderboard (0.80/1.0) via LoRA (r=32, ~888M trainable parameters) with 20+ single-variable ablations validating deployment-critical training decisions. Core skills include PyTorch, Hugging Face Transformers, PEFT/LoRA, FP16 mixed precision, CUDA debugging, and Triton - all directly applicable to edge AI model optimization and hardware-aware deployment. Available immediately for a European internship; based in France with no work authorization barriers.

CORE COMPETENCIES

Edge AI Deployment | Model Optimization | Hardware-Aware Inference | ML Pipeline Engineering | Model Fine-Tuning | Ablation Studies | Mixed Precision Training | Production AI Deployment

WORK EXPERIENCE

Orange Maroc Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered an end-to-end ML pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** (upload speed, latency, jitter), deploying output directly to the network optimization team.
- Designed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, enabling automated production deployment of maintenance recommendations without manual analyst intervention.
- Reduced manual network analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and reproducible Scikit-learn pipeline components.
- Delivered stakeholder presentations translating model outputs into actionable maintenance decisions, supported by **Power BI** dashboards integrating classification results with live network metrics.

PROJECTS

NVIDIA Nemotron Reasoning Challenge Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** on a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model using LoRA (r=32, ~888M trainable parameters) with SFTTrainer completion-only loss over 7,828 verified chain-of-thought examples.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** - including monkey-patched allocator warmup, stubbed Mamba/Cutlass imports, disabled fast-path CUDA kernels, fixed Triton ptxas permissions, and replaced rmsnorm_fn with a pure PyTorch fallback - all in strict dependency order; validated decisions through 20+ single-variable ablations confirming packing, synthetic data, and adapter size as critical deployment-affecting variables.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st on the EURECOM private Kaggle class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and FP16 mixed precision training across 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; confirmed that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC) via ablation, demonstrating hardware-efficient model deployment through 10-view TTA ensemble.

PyTorch · OpenCLIP · ViT · Kaggle T4 GPU · FP16 Mixed Precision

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Image Security, Statistics, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya 2021 - 2023

SKILLS

ML Frameworks & Edge Engineering: PyTorch | Hugging Face Transformers | PEFT / LoRA | Unsloth / TRL | Scikit-learn | FP16/BF16 Mixed Precision | vLLM

Model Deployment & MLOps: CUDA Debugging | Triton | Ablation Studies | Feature Engineering | Git | Linux | Docker (basic)

Data & Visualization: Python | Pandas | NumPy | SQL | Power BI | Matplotlib | Seaborn